

12. odvisnost mehične vode od tlaka

Priprejamo delovanje Elara depozitirane vode.

V kudi je voda batera depozitirana in vrtomils najo inno monometen in temometen - rogneli bomo na roko temperatur - tlak bo narastel - spiziculi bomo voda reu... voda bo opet ranelu. $(3-5)^{\circ}\text{C}$

8 meritev ro tlak pri mehični

merit: voda vabo in prisine

$T[^{\circ}\text{C}]$	$p[\text{Bar}]$
99	0,9
95	0,7
92	0,65
89	0,65
86	0,55
83	0,45
80	0,40
77	0,35
74	0,30

~~Ustrezno tlak in temperatura~~

$$T_0 = 22^{\circ}\text{C} \pm 1^{\circ}\text{C}$$

12/01/10

Člaus slovenova:

$$\ln\left(\frac{P}{P_0}\right) = \frac{M g_i}{R} \left(\frac{1}{T_0} - \frac{1}{T} \right)$$

$$\ln\left(\frac{P}{P_0}\right) = \frac{M g_i}{R} \frac{1}{T_0} - \frac{M g_i}{R} \frac{1}{T}$$

$\underbrace{\quad}_Y \quad \underbrace{\quad}_{+m} \quad \underbrace{\quad}_k \quad \underbrace{\quad}_X$

iz fitovanja:

$$k = -5466 \pm 130 =$$

$$m = 14,58 \pm 0,35$$

$$M = 18,01 \frac{\text{g}}{\text{mol}}$$

$$g_i = 2,256 \cdot 10^6 \frac{\text{J}}{\text{kg}}$$

$$R = 8,314 \frac{\text{J}}{\text{mol K}}$$

$$k_{\text{calc}} = - \frac{M g_i}{R} = - \frac{0,01801 \cdot \frac{\text{kg}}{\text{mol}} \cdot 2,256 \cdot 10^6 \frac{\text{J}}{\text{kg}}}{8,314 \frac{\text{J}}{\text{mol K}}} = -4886,94$$

$$m_{\text{calc}} = \frac{M g_i}{R} \frac{1}{T_0} = -k_{\text{calc}} \cdot \frac{1}{T_0} = +4886,94 \cdot 0,00288 = 14,07$$

$$m_{\text{calc}} =$$

grafi